

James J Nadeau

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Skills

- 20 years of experience servicing and developing software and data systems, 10 years consulting happy clients.
- Lead teams over several years with up to 5 team members. Managing developers and support staff for over 10 years
- Proven experience securing, upgrading and managing legacy software services.
- Enterprise Level Data Systems Architecture using Databricks, Plotly, PowerBI and Python/R
- Cloud Scale Web Systems and API's hosted on AWS, Google Cloud, and Azure
- Languages/Frameworks: Python/Spark, Javascript/[Node.js](#), Bash, PHP, SQL, XML, Vue, React, Svelte
- Technologies: Linux, Docker, MySQL, Postgres, SQL Server, AWS, Azure, Google Cloud, Redis, Docker, CI/CD, CRM, ERP, REST, GraphQL

Domain Expertise & Project Highlights

- **Education** - Helped the state of Vermont as a consultant architect and implemented plans to first handle a csv based collection system whose project went awry. I then joined the state and helped them move to a Ed-Fi based JSON REST API to collect data state wide. The first production collection of over 14 billion change records was completed this September.
- **CRM** - Created CRM applications focused on the HVAC industry for over 150 local businesses and brands such as Bosch, Fujitsu, Rinnai. Worked on non-profit and large enterprise CRM systems for clients and created custom integrations for things like Quickbooks and Stripe.
- **Finance** - Created tools for financial advisors to choose portfolio packages for Verisigh Inc. Worked on tax and financial calculations relating to state wide taxes and education finance.
- **Healthcare** - Modified existing FastAPI based application to integrate telemedicine and medical record keeping systems used across 10 states. Turned a 1 month consulting project into a 6 month enhancement plan for the client.
- **DevOps CI/CD, Docker and Kubernetes** - Helped Clevertex create CI/CD pipelines and boilerplate configuration used to deploy via Jenkins and Kubernetes. Created CI/CD pipelines for Python FastAPI and Flask applications deployed through docker. Deployed .Net applications through Docker Compose on Azure. 10 years of DevOps experience large scale production applications used.
- **Databricks, Jupyter, Large Scale Data Processing and Quality Control** - 2 years experience setting up and leading a migration to a Databricks based environment processing millions of rows per day. Created streaming pipelines from API based data used by a team of 25 analysts and developers to inform business and public operations

Work Experience

2020-Present

Vermont Agency of Education, Education Data Administrator.

Lead the agency's data technology initiatives. Led a team of 4:

- Automated workloads and made information more readily available by creating a public facing knowledge base. Cut email questions and unique identification process time by 50% for the State Longitudinal Data System support group.
- Created and administrated a python based Jupyter Hub environment to manage Jupyter environments across the division. The system was Integrated with Active Directory and used Kerberos tokens to authenticate.
- Migrated Jupyter Hub system to Databricks to comply with security and logging requirements. Migrated Cron based scripts to Databricks Jobs & Pipelines. Created a streaming pipeline to deal with 54 million rows of data.
- Calculated key elements of education data used to determine state education taxes. Ensured accuracy and quality.
- Worked with 52 local education agencies across 4 different student information system providers to ensure a smooth transition between data collection systems. Provide support to data quality issues from data managers.
- Advised Agency of Digital Services on docker deployments, Kubernetes and Apache Airflow deployments for Ed-Fi and other tools.

2018-2020

Competitive Computing (C2), Software Engineer & Tech Lead

Worked on the Agency of Education (AOE) Enterprise Data Environment Project. Architected Jupyter/Python based environment to help AOE get their work done faster, in a unified, self documenting literate code development tool.

- Through the use of systems I helped create, 1,389 hours of manual work each year and over 40,000 touch points removed from data processing efforts. The contract was extended 3 times due to the client creating more work.
- Converted legacy Oracle sql procedures that no AOE/ADS staff knew how to maintain to Jupyter/Python. Reducing

risk for the organization and allowing Oracle to be retired.

- Provided advice for various technological challenges presented to the team.
- Represented AOE interests from a technical perspective for various projects and resources.
- Mentored AOE staff on the use of Python and coding best practices.

2014-2018

Clevertch.biz, Senior Developer and Project Manager

Created applications and websites for clients, advised architecture choices, and helped troubleshoot escalated issues. Created new project boilerplate based on NodeJS, Redis and Docker deploying to Kubernetes.

- Worked to create version 2.0 of the lookbooks application in use by Getty Image's artists. I contributed major parts to the upgrade including adding drag and drop asset management, an HTML5 based uploader, a migration to a template based front end UI instead of the existing jQuery.
- Increased the page load speed client side on several projects. Added external integrations to existing applications.
- Used a Docker based infrastructure to deploy scalable API's and backend processors capable of sending millions of custom generated emails per day. Used event driven architecture to achieve scale and cost savings.
- Implemented a Docker/Jenkins/ESLint based project boilerplate that is now used on all new projects, deployed to Kubernetes via CI/CD. Provided oversight on DevOps operations and escalations.
- Provided technical leadership to various projects ranging from ordering AI driven taxis to using XBOX Kinect peripherals to do package size/volume estimates in near real time for sorting systems..

2013-2014

Software Contractor, for MJB Consulting and Wheelstv.com

Went into business for myself and developed custom solutions for my customers.

- Created video SEO tool that would add pre/post rolls and overlays to a set of car promotion videos, then upload them to youtube/facebook/vimeo through a website/dashboard. Also created an encoding to output various formats.
- Provided Department of Labor form 5500 Data in a consolidated database that allows quick searching and comparison benchmarking.
- Designed and developed 401K portfolio research and creation tool for Verisight Inc. that generated PDF proposals for its customers. Also worked with DOL form 5500 data analysis and conversion to an REST API

2011- 2013

Red Barn Media Group, Senior Developer and Systems Architect,

Made all technical decisions and provided security review for 4 employees.

- Provided technical consultation and estimates on all projects. Handled any technical escalations.
- Created Apache Cordova and jQuery Mobile based Mobile applications using Camera QR code capture
- Integrated with external companies and services including:Brainshark, Twillio, SAML, Google Maps, Mailgun
- Added browser based phone support into our systems, immediately converting incoming calls to support tickets and allowed support team tracking. Saved countless hours per week
- Created real time analytics of system health and marketing targets. Implemented Invoicing System that all accounting now goes through.
- Made Custom Drupal Modules and deployment system to bridge gap between desired system functionality and Drupal Core.

2008- 2011

National Gardening Association, IT Administrator/Systems Architect - garden.org, kidsgardening.org, others....

In charge of all IT related tasks for the organization. Administered Internal Network, Mail Server, External Connections, and Internal VOIP network. Started and promoted Facebook page to 10,000 users within 3 months.

- Administered many Linux Servers of various flavors and vintage; serving over 5 million web pages per year for internal and external large, demanding clients: Home Depot, Scotts, Subaru, Yum, Growers Supply, FDA, USDA...
- Created a load balanced smtp cluster for the bulk sending of over 2 million email newsletters per month.
- Migrated content and applications over to Drupal based system involving several developers and designers.

2003-2007

RPI Distance Education Department, Admin/Support

Helped establish RPI as a leader in distance education. Administered linux and windows based servers. Worked with other departments for infrastructure set up and problems. Answered calls for support line. Helped design and create web pages, taught others to do the same. I had a nice office on campus as a undergrad!

Education

2002-2007 Rensselaer Polytechnic Institute(RPI), Troy, NY

Computer and Systems Engineering, Computer Science

